



Chapter 13: Differentiation of Trigonometric, Exponential and Logarithmic Functions

Name: _____ ()

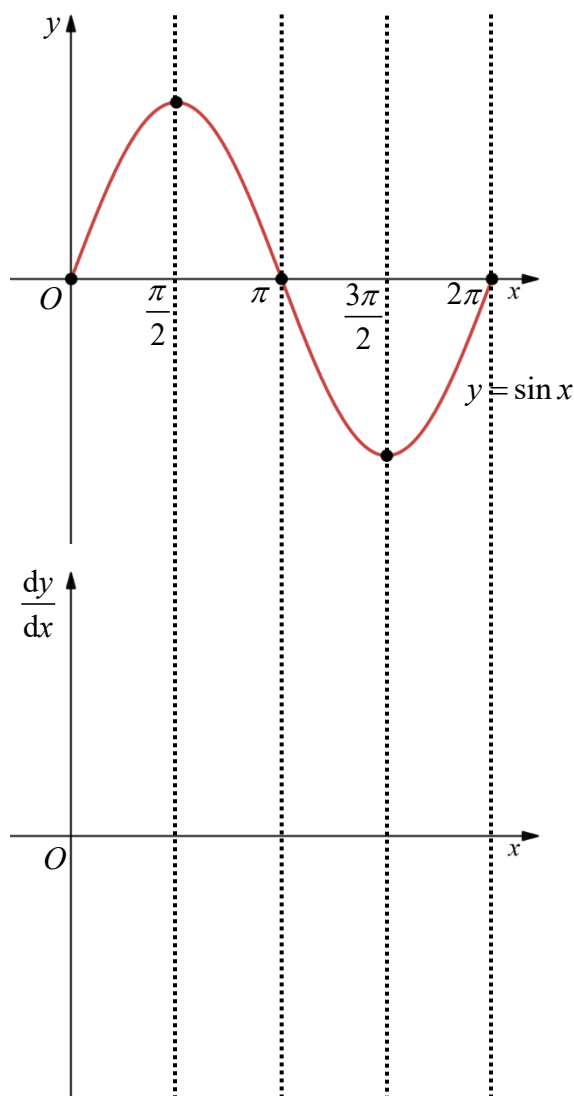
Class: _____

13.1 Derivatives of Trigonometric Functions

(A) Derivatives of $\sin x$, $\cos x$ and $\tan x$.

Discussion: The following shows the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$.

Sketch the corresponding graph of its derivative function $y = \frac{d}{dx} \sin x$ for $0 \leq x \leq 2\pi$. (hint: estimate the gradient for each of the dotted points. Or use this link: <https://www.geogebra.org/m/YhVNnJnJ>)

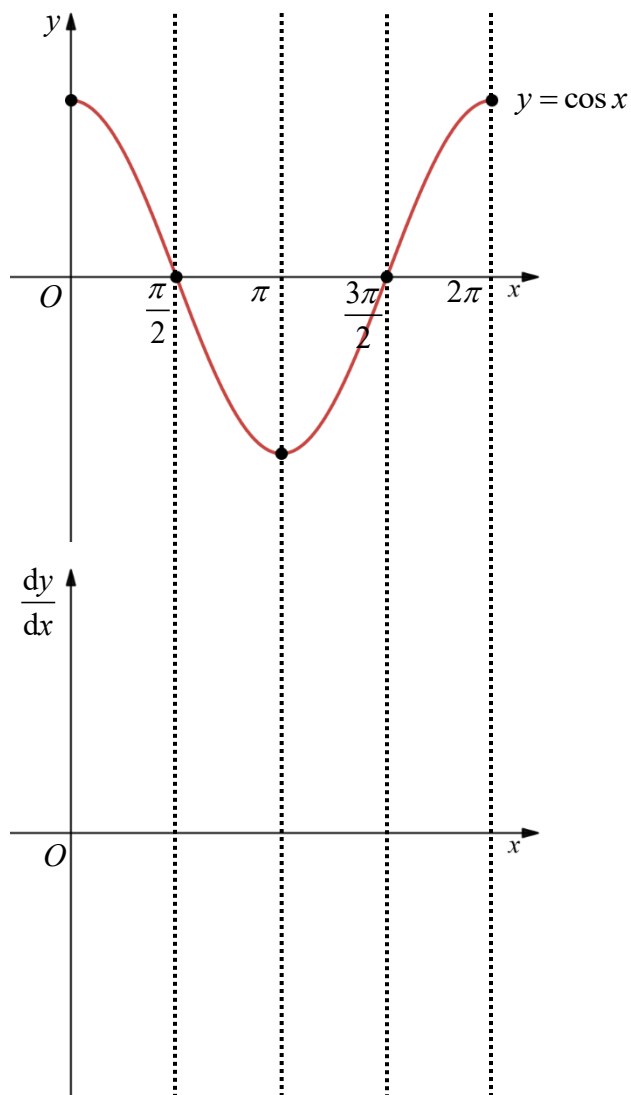


$$\frac{d}{dx}(\sin x) =$$

where x is in radians

Discussion: The following shows the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$.

Sketch the corresponding graph of its derivative function $y = \frac{d}{dx} \cos x$ for $0 \leq x \leq 2\pi$. (hint: estimate the gradient for each of the dotted points. Or use this link: <https://www.geogebra.org/m/YhVNnJnJ>)



$$\frac{d}{dx}(\cos x) =$$

where x is in radians

Discussion: show that $\frac{d}{dx}(\tan x) = \sec^2 x$. (hint: let $\tan x = \frac{\sin x}{\cos x}$ and apply the quotient rule)

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

where x is in radians



Differentiate each of the following with respect to x , where x is in radians.

(a) $4 \sin x$	(b) $y = 3 \tan x - 2 \sin x$
(c) $y = \frac{\tan x}{2x+3}$	(d) $y = \frac{2 \cos x}{5x+1}$
(e) $2x^2 \cos x$	(f) $3x^5 \sin x$
(g) $f(x) = (3x + 2 \sin x)^4$	(h) $f(x) = (3x + 5 \cos x)^6$

(B) Derivatives of $\sin f(x)$, $\cos f(x)$ and $\tan f(x)$.

How do we differentiate $y = \sin(2x+1)$? The function $y = \sin(2x+1)$ is called a composite function (recall chapter 11). Hence to differentiate it, we need to apply the chain rule.

$$\text{Let } y = \sin u \quad \text{and} \quad u = f(x) = 2x+1,$$

$$\text{i.e. } \frac{dy}{du} = \cos u \quad \text{and} \quad \frac{du}{dx} = f'(x) = 2.$$

$$\begin{aligned} \text{Applying the Chain Rule, } \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\ &= \cos u \times 2 \\ &= 2 \cos(2x+1) \end{aligned}$$

In general,

$$\begin{aligned} \frac{d}{dx} [\sin f(x)] &= f'(x) \cos f(x), \\ \frac{d}{dx} [\cos f(x)] &= -f'(x) \sin f(x), \\ \frac{d}{dx} [\tan f(x)] &= f'(x) \sec^2 f(x), \text{ where } x \text{ is in radians} \end{aligned}$$

Hence, if $f(x)$ is a linear function, i.e. $f(x) = ax+b$, where a and b are constants, then

$$\begin{aligned} \frac{d}{dx} [\sin(ax+b)] &= a \cos(ax+b), \\ \frac{d}{dx} [\cos(ax+b)] &= -a \sin(ax+b), \\ \frac{d}{dx} [\tan(ax+b)] &= a \sec^2(ax+b), \text{ where } x \text{ is in radians} \end{aligned}$$



Differentiate each of the following with respect to x , where x is in radians.

(a) $2 \sin\left(3x + \frac{\pi}{2}\right)$

(b) $y = 5 \cos\left(2x + \frac{\pi}{4}\right)$

(c) $y = 3x^2 \tan 5x$

(d) $y = 4x \tan\left(3x - \frac{\pi}{3}\right)$

(e) $f(x) = \frac{\sin 4x}{\cos\left(\frac{\pi}{2} - 2x\right)}$

$$(f) \ f(x) = \frac{2 \sin 5x}{\cos\left(\frac{\pi}{2} - 4x\right)}$$

(C) Derivatives of $\sin f''(x)$, $\cos f''(x)$ and $\tan f''(x)$.



Differentiate each of the following with respect to x , where x is in radians.

$$(a) \ y = 5 \sin^3 2x$$

$$(b) \ y = 3 \cos^4\left(\frac{5x}{9}\right)$$

(c) $y = 2 \cos(\pi x + 1)^3 + 4 \tan^3\left(\frac{x}{2} + \frac{\pi}{3}\right)$

(d) $y = 2 \sin^5\left(\frac{x}{\sqrt{3}}\right) - 5 \tan^2\left(2x + \frac{\pi}{2}\right)$

13.2 Derivatives of Exponential Functions

- 1) The derivative of e^x is given by:

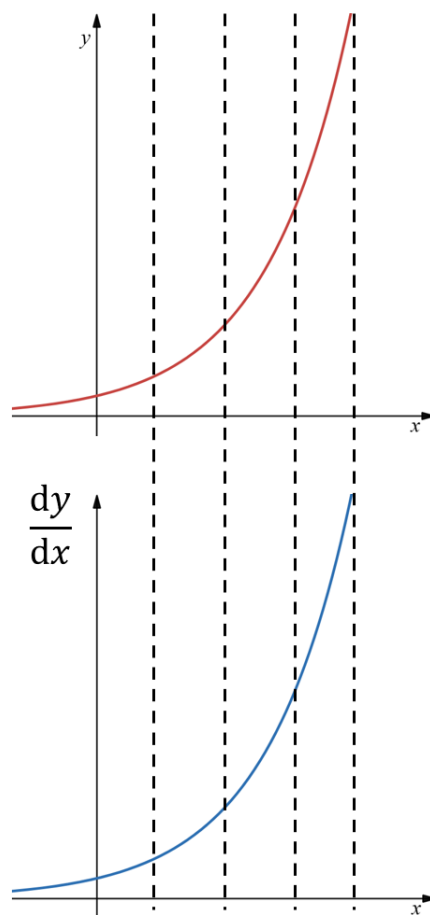
$$\frac{d}{dx}(e^x) = e^x$$

- 2) Similar to page 4, we will apply the chain rule to find the derivative of $e^{f(x)}$:

$$\frac{d}{dx}[e^{f(x)}] = e^{f(x)} \times f'(x)$$

For example, if $f(x)$ is a linear function, i.e. $f(x) = ax + b$, where a and b are constants, then

$$\frac{d}{dx}[e^{ax+b}] = ae^{ax+b}$$



Differentiate each of the following with respect to x .

(a) e^{3x-4}

(b) $3e^{5x}$

(c) $3x^2e^{2x}$

(d) $f(x) = e^{3x^2-5x}$

(e) $y = e^{2x+1} \sin 2x$, where x is in radians	(f) $y = e^{5+2x} \cos 2x$, where x is in radians
(g) $f(x) = \frac{e^{3-4x}}{x}$	(h) $f(x) = \frac{e^{4-x^3}}{6e}$



Given that $y = (e^x + e^{2x})^4$, find the value of $\frac{dy}{dx}$ at the point where $x = 0$.

13.3 Derivatives of Logarithmic Functions

- 1) The derivative of $\ln x$ is given by:

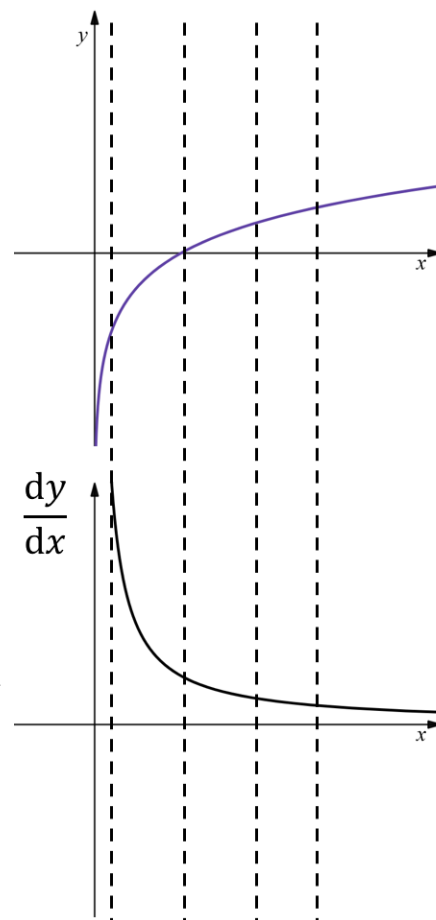
$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

- 2) Similar to page 4, we will apply the chain rule to find the derivative of $\ln f(x)$:

$$\frac{d}{dx}[\ln f(x)] = \frac{1}{f(x)} \times f'(x)$$

For example, if $f(x)$ is a linear function, i.e. $f(x) = ax + b$, where a and b are constants, then

$$\frac{d}{dx}[e^{ax+b}] = ae^{ax+b}$$



EXAMPLE Differentiate each of the following with respect to x .

(a) $\ln(5x + 2)$	(b) $\ln(5x^2 + 7x)$
(c) $e^y = \cos 3x$, where x is in radians	(d) $e^y = \sin 4x$, where x is in radians

(e) $f(x) = \ln \sqrt{\frac{3+x}{3-x}}$

Method 1

(f) $f(x) = \ln \sqrt{\frac{3+x}{3-x}}$

Method 2

(g) $f(x) = \ln \sqrt[3]{x^3 + 2x}$



1. Given that $y = x \ln (\tan 2x)$, where x is in radians, find the value of $\frac{dy}{dx}$ when $x = 0.5$.
2. Given that $f(x) = \frac{\ln 2x}{1+x}$, where $x > 0$, find the value of $f'(1.2)$.
3. Find the first derivative of $[\ln (ex + 6)]^3$ with respect to x when $x = e$, leaving your answer in terms of e .

13.4 Further Application of Differentiation

Now that we have learned how to differentiate trigonometric, exponential and logarithmic functions, we will learn how we can apply these concepts.

Equation of tangent and normal



Find the equations of the tangent and the normal to the curve $y = 4x + \cos 2x$ at the point where $x = \frac{\pi}{12}$.



The equation of the curve is $y = \ln(2x^2 + 5x)$, where $x > 0$. Find the x -coordinate of the point on the curve at which the normal to the curve is perpendicular to the line $6y = 2x + 25$.

Connected rate of change



Given that $y = \ln\left(\frac{3x-4}{2x+5}\right)$, find

- (i) $\frac{dy}{dx}$,
- (ii) the rate of change of x when $x = 2$, given that y is changing at a rate of 2.5 units per second.



Given that $y = 1 + 2 \sin^2 x$ for $0 \leq x \leq \pi$ and that x is increasing at a rate of 0.2 radian per second, find the rate of change of y with respect to time when $x = \frac{\pi}{3}$.

Stationary points and their nature



Given that $y = 3x \ln x$, find $\frac{dy}{dx}$. Hence, determine the coordinates of the stationary point of the curve $y = 3x \ln x$.



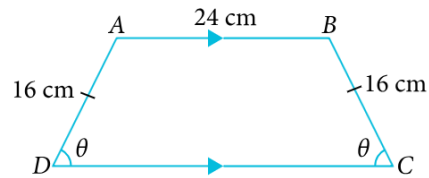
Given that $y = e^x \sin 2x$, find the value of x where $0 \leq x \leq \frac{\pi}{2}$ for which y has a stationary value. Determine the nature of this stationary value.

Maxima/Minima Problems



In the diagram, $ABCD$ is a trapezium such that $\angle ADC = \angle BCD = \theta$ radians, AB is parallel to DC , $AB = 24$ cm and $AD = BC = 16$ cm.

- (i) Show that the area of the trapezium, S cm², is given by $S = 128 \sin \theta (3 + 2 \cos \theta)$.
- (ii) Hence, find the value of θ for which S has a stationary value.
- (iii) Determine whether this value of θ makes S a maximum or a minimum.



Answers

Example 1

- (a) $4 \cos x$

$\varnothing$

- (b) $3 \sec^2 x - 2 \cos x$

$\varnothing$

- (c) $\frac{(2x+3) \sec^2 x - 2 \tan x}{(2x+3)^2}$

$\varnothing$

- (d) $\frac{-2(5x+1) \sin x - 10 \cos x}{(5x+1)^2}$

$\varnothing$

- (e) $4x \cos x - 2x^2 \sin x$

$\varnothing$

- (f) $15x^4 \sin x + 3x^5 \cos x$

$\varnothing$

- (g) $4(3x + 2 \sin x)^3(3 + 2 \cos x)$

$\varnothing$

- (h) $6(3x + 5 \cos x)^5(3 - 5 \sin x)$

$\varnothing$

Example 2

- (a) $6 \cos(3x + \frac{\pi}{2})$

$\varnothing$

- (b) $-10 \sin(2x + \frac{\pi}{4})$

$\varnothing$

- (c) $6x \tan(5x) + 15x^2 \sec^2(5x)$

$\varnothing$

- (d) $4 \tan(3x - \frac{\pi}{3}) + 12x \sec^2(3x - \frac{\pi}{3})$

$\varnothing$

- (e) $-4 \sin 2x$

$\varnothing$

- (f) $\frac{10 \cos(5x) \sin(4x) - 8 \sin(5x) \cos(4x)}{\sin^2(4x)}$

$\varnothing$

Example 3

- (a) $30 \sin^2(2x) \cos(2x)$

$\varnothing$

- (b) $-\frac{20}{3} \cos^3(\frac{5x}{9}) \sin(\frac{5x}{9})$

$\varnothing$

- (c) $-6\pi(\pi x + 1)^2 \sin((\pi x + 1)^3) + 6 \tan^2(\frac{x}{2} + \frac{\pi}{3}) \sec^2(\frac{x}{2} + \frac{\pi}{3})$

$\varnothing$

- (d) $\frac{10}{\sqrt{3}} \sin^4(\frac{x}{\sqrt{3}}) \cos(\frac{x}{\sqrt{3}}) - 20 \tan(2x + \frac{\pi}{2}) \sec^2(2x + \frac{\pi}{2})$

$\varnothing$

Example 4

- (a) $3e^{3x-4}$

$\varnothing$

- (b) $15e^{5x}$

$\varnothing$

- (c) $6xe^{2x}(1+x)$

$\varnothing$

- (d) $(6x-5)e^{3x^2-5x}$

$\varnothing$

- (e) $2e^{2x+1}(\sin 2x + \cos 2x)$

$\varnothing$

- (f) $2e^{5+2x}(\cos 2x - \sin 2x)$

$\varnothing$

- (g) $\frac{-e^{3-4x}(4x+1)}{x^2}$

$\varnothing$

- (h) $-\frac{x^2}{2}e^{3-x^3}$

Example 5

96

Example 6

- (a) $\frac{5}{5x+2}$

[⌕](#)

- (b) $\frac{10x+7}{5x^2+7x}$

[⌕](#)

- (c) $-3 \tan 3x$

[⌕](#)

- (d) $4 \cot 4x$

[⌕](#)

- (e) $\frac{3}{9-x^2}$

[⌕](#)

- (f) $\frac{3}{9-x^2}$

[⌕](#)

- (g) $\frac{3x^2+2}{3(x^3+2x)}$

[⌕](#)

Example 7

- Example 1: $\ln(\tan 1) + \frac{2}{\sin 2}$ (or ≈ 4.84) [⌕](#)

- Example 2: $\frac{11-6 \ln 2.4}{29.04}$ (or ≈ 0.198) [⌕](#)

- Example 3: $\frac{3e}{e^2+6} [\ln(e^2+6)]^2$

[⌕](#)

Example 8

Tangent: $y = 3x + \frac{\pi}{12} + \frac{\sqrt{3}}{2}$, Normal: $y = -\frac{1}{3}x + \frac{13\pi}{36} + \frac{\sqrt{3}}{2}$

Example 9

$$x = 5$$

Example 10

(i) $\frac{23}{(3x-4)(2x+5)}$, (ii) $\frac{45}{23}$ (or ≈ 1.96) units per second [⌕](#)

Example 11

$\frac{\sqrt{3}}{5}$ (or ≈ 0.346) units per second [⌕](#)

Example 12

$$\left(\frac{1}{e}, -\frac{3}{e}\right)$$

Example 13

$x = 1.02$, y is a maximum

Example 14

$\theta = 1.13$, S is maximum